

Project Outline

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Project Title: Locally self-supervised deep learning for vision

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Project Description

The aim of this project is to develop a biologically inspired approach to deep learning for computer vision. The core premise of this project is that biological neural systems do not learn through global error backpropagation, like artificial neural networks (ANNs) do, but instead on local, self-supervised learning rules.

The main substance of the project is the design, implementation, and analysis of a locally self-supervised learning algorithm. This algorithm will operate on the principle of learning compact, efficient encodings of local input patterns (i.e. from a small part of an image) in an unsupervised manner, taking inspiration from models such as Variational Autoencoders (VAEs). These local encoders will be applied in a convolutional/hierarchical manner across an image (like CNNs without backpropagation), allowing the network to build representations of visual data across larger inputs without requiring externally provided labels for each layer. The algorithm is therefore designed to be flexible, allowing for experimentation with various network structures, activation functions, and normalization techniques.

The project will explore how choices in network structure, activation functions, normalization techniques etc. affect the performance and quality of the algorithm. The essential aspects making this project worthwhile are its potential to contribute to more efficient, robust and biological or natural (brain-like) machine learning models. By moving away from backpropagation, we can explore systems that could learn continuously from streaming data (mimicking a brain, resulting in better flexibility and efficiency) and demonstrate better generalization from a more limited number of examples (more robustness). It learns from raw data by itself, not externally given data-sets, which is a huge advantage as it doesn't require user input and datasets are expensive.

The end-goals are:

A functional and optimised software implementation of the proposed algorithm.

A thorough evaluation of its performance on benchmark vision tasks (e.g. image classification, reconstruction etc.) compared to more commonly used conventional methods.

Identification (or a foundation for the investigation) of promising architectural variants, applications, and extensions of the core machine learning algorithms for further research and improvements to be made.

Proposed Tasks

1. Conduct deeper research and strengthen/broaden knowledge of related topics
 - a. Self-supervised and unsupervised learning in deep neural networks (and computer vision especially)
 - b. Variable autoencoders and similar models
 - c. Biological and local learning methods
 - d. Existing locally trained neural networks
 - e. Alternatives to backpropagation
2. Learning and choosing the technology and frameworks that I will be building the project upon (tensorflow/pytorch etc.)
 - a. Performance requirements and optimisation considerations
 - b. Figuring out their best use cases
 - c. Learning how to optimally use them through tutorials
3. Implementing the algorithm using chosen frameworks successfully
 - a. Start with making sure it's successful with a smaller task
 - b. Small encoder/decoder modules
 - c. Local training objectives
 - d. Convolutional/hierarchical composition
4. Experiment with different network configurations (architecture, parameters etc.)
 - a. Size of local modules
 - b. Activation functions
 - c. Normalisation/regularisation techniques
5. Evaluate outcome and potential improvements or extensions in a final report
 - a. Assess quality and performance
 - b. Identify where improvements can be made and if possible, apply them
 - c. Investigate potential extensions (i.e. different areas where ideas can be applied such as robotics)

Project Deliverables

- A working implementation of the locally self-supervised vision algorithm
- Source code and documentation (final project outline, timeline/journal, key papers reviewed, README on how to install/run and build upon the source etc.) hosted in a repository such as GitHub
- A collection of outputs from key experiments e.g. important trained model checkpoints, scripts for reproduction of training/evaluation runs, logs of training dynamics, visual representations etc.
- Final Report: Comprehensive document detailing the project's motivation, methodology, implementation, results, conclusion etc.

Initial Annotated Bibliography

- Tensorflow/PyTorch Official Documentation and Tutorials
 - Essential for guidance in the practical implementation of the project
 - <https://www.tensorflow.org/tutorials>
- Backpropagation and the brain (Lillicrap, Santoro, Marris, Akerman, Hinton [2020])
 - Discusses backpropagation in the brain and whether or not backpropagation offers insights for understanding learning in the cortex
 - <https://pubmed.ncbi.nlm.nih.gov/32303713/>
- Computational Cognitive Neuroscience (O'Reilly, Munakata, Frank, Hazy, Contributors [2012])
 - Discusses foundations of biological neural network models and machine learning rules
 - https://www.researchgate.net/profile/Randall-Oreilly/publication/233822187_Computational_Exploration_in_Cognitive_Neuroscience/links/606e18dd92851c8a7baf2f98/Computational-Exploration-in-Cognitive-Neuroscience.pdf
- Deep predictive coding networks for video prediction and unsupervised learning (Lotter, Kreiman, Cox [2016])
 - Discusses solving unsupervised learning tasks with deep learning algorithms, inspired by the concept of “predictive coding” from neuroscience literature
 - <https://arxiv.org/abs/1605.08104>
- Auto-Encoding Variational Bayes (Kingma, Welling [2013])
 - Discusses learning both inference and generation in latent variable models using neural networks, making variational inference practical for large-scale deep learning
 - <https://arxiv.org/abs/1312.6114>